

1. Administrative & Study Identification

Full Study Title: A Phase 2, Randomized, Double-Blind, Placebo-Controlled Study of V940 in Combination With Pembrolizumab and Chemotherapy as First-Line Treatment for Participants With Metastatic Squamous NSCLC **Short Title/Acronym:** INTERpath-013 (V940-013) **Protocol Number:** V940-013 **NCT Number:** NCT07221474 **Sponsor Name:** Merck Sharp & Dohme LLC (in collaboration with Moderna) **Investigational Product:** V940 (Intismeran autogene) in combination with Pembrolizumab and Platinum-Doublet Chemotherapy **Phase of Development:** Phase 2 **Medical Monitor:** Merck Sharp & Dohme Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate whether V940 in combination with pembrolizumab and chemotherapy improves Progression-Free Survival (PFS) and Overall Survival (OS) compared to placebo combined with pembrolizumab and chemotherapy. **Secondary Objectives:** To evaluate Overall Response Rate (ORR), Duration of Response (DOR), and the Number of Participants who Experience an Adverse Event (AE).

2.2 Background & Rationale To address the need for improved first-line therapies for patients with metastatic squamous non-small cell lung cancer (NSCLC). V940 is an individualized mRNA-based neoantigen therapy (personalized cancer vaccine) designed to help a patient's immune system target specific cancer mutations. This study aims to determine if priming the immune system with V940 enhances the clinical efficacy of standard-of-care immunotherapy (pembrolizumab) and chemotherapy.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized (Parallel Assignment) **Trial Status:** Not Yet Recruiting

3.2 Planned Enrollment & Duration Target \$N\$: 180 participants **Number of Sites:** Global multicenter trial **Estimated Study Start:** 12-Dec-2025 **Estimated Primary Completion:** 02-Jul-2029

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age $\geq$ 18 years. 2. Histologically or cytologically confirmed diagnosis of squamous non-small cell lung cancer (NSCLC) (Stage IV: M1a, M1b, M1c1, M1c2 per AJCC Staging Manual, Version 9). Mixed tumors will be characterized by the predominant cell type; small cell elements are not permitted. 3. Has measurable disease per RECIST 1.1 as assessed by the local site investigator/radiology. 4. Has provided a tissue sample collected at or after the diagnosis of metastatic disease AND from a site not previously irradiated. 5. Eastern Cooperative Oncology Group (ECOG) performance status of 0 to 1 assessed within 7 days before randomization. 6. Has a life expectancy of at least 3 months and adequate organ function. 7. Adverse events (AEs) due to previous anticancer therapies must have recovered to $\leq$ Grade 1 (endocrine-related AEs adequately treated with hormone replacement or $\leq$ Grade 2 neuropathy are eligible). 8. HIV-infected participants must have well-controlled HIV on antiretroviral therapy (ART). HBV (HBsAg positive) and HCV participants are eligible if treated with antiviral therapy for at least 4 weeks with an undetectable viral load prior to randomization.

4.2 Exclusion Criteria 1. Has received prior systemic anticancer therapy for their metastatic NSCLC. 2. Has received prior treatment with a cancer vaccine, including another personalized cancer vaccine (PCV). 3. Has received prior therapy with an anti-PD-1, anti-PD-L1, or anti-PD-L2 agent (neoadjuvant/adjuvant setting for nonmetastatic NSCLC allowed if completed at least 12 months before diagnosis of metastatic NSCLC). 4. Known active central nervous system (CNS) metastases and/or carcinomatous meningitis. 5. Known additional malignancy that is progressing or has required active treatment within the past 3 years. 6. Is HIV-infected with a history of Kaposi's sarcoma and/or Multicentric Castleman's Disease. 7. Has received prior radiotherapy within 2 weeks of start of study intervention, requires corticosteroids for radiation-related toxicities, or has received radiation therapy to the lung >30 gray within 6 months. 8. Severe hypersensitivity ($\geq$ Grade 3) to V940, pembrolizumab, or allowed chemotherapy agents (e.g., carboplatin/Paraplatin) and/or any of their excipients. 9. Active autoimmune disease requiring systemic treatment in the past 2 years, or a history of pneumonitis/interstitial lung disease requiring steroids (or current). 10. Active infection requiring systemic therapy, history of stem cell/solid organ transplant, or has not adequately recovered from major surgery (or has ongoing surgical complications). 11. Has received an investigational agent/device within 4 weeks prior to intervention. 12. Diagnosis of immunodeficiency, or receiving chronic systemic steroid therapy (>10 mg daily prednisone equivalent) or any other immunosuppressive therapy within 7 days prior to the first dose.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: - Induction:

Pembrolizumab 400 mg IV Q6W + Chemotherapy (Carboplatin [Paraplatin] + Paclitaxel or Nab-paclitaxel) + V940 1 mg IM (or Placebo) administered on Days 1 and 22 of Cycle 2 (Q3W) for up to 2 doses. - **Maintenance:** Pembrolizumab 400 mg IV Q6W for up to 15 doses. **Route of Administration:** Intramuscular (IM) for V940/Placebo; Intravenous (IV) for Pembrolizumab and Chemotherapy. **Duration of Treatment:** Up to 15 doses of Pembrolizumab maintenance following induction.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for chemotherapy and immunotherapy-related side effects. Killed vaccines are allowed. **Prohibited:** Live or live-attenuated vaccines within 30 days before the first dose, chronic systemic steroid therapy (>10 mg daily prednisone equivalent), or other investigational agents within 4 weeks.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor tissue collection, Baseline Imaging
Induction	Cycles 1 & 2	Pembrolizumab + Chemotherapy IV infusion, V940/Placebo IM Injection (Cycle 2), Safety Labs
Maintenance	Q6W	Pembrolizumab IV infusion, Efficacy Assessment (Imaging), AE monitoring
End of Study	Upon PD	Final Vital Signs, Exit Interview, Follow-up for Overall Survival

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints - Progression-Free Survival (PFS): Defined as the time from randomization to the first documented progressive disease (PD) or death due to any cause. Assessed by Blinded Independent Central Review (BICR) per RECIST 1.1. PD is defined as $\geq 20\%$ increase in the sum of diameters of target lesions along with an absolute increase of ≥ 5 mm, or the appearance of new lesions. - **Overall Survival (OS):** Defined as the time from randomization to death due to any cause.

7.2 Secondary Endpoints - Overall Response Rate (ORR) - Duration of Response (DOR) - Number of Participants who Experience an Adverse Event (AE)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via regular clinical visits, physical examinations, and laboratory panels to

determine the number of participants experiencing AEs. **Serious Adverse Events (SAEs):** Required reporting within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** External Data Monitoring Committee assigned to review ongoing safety and efficacy signals.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy endpoints (PFS/OS). Safety Set for all AE/SAE evaluations. **Sample Size Justification:** \$N=180\$ powered to detect statistically meaningful improvements in PFS and OS when adding V940 to standard of care. **Handling of Missing Data:** Standard Kaplan-Meier survival estimation censoring rules applied for time-to-event outcomes.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring visits to ensure protocol and regulatory compliance. **Audit Trail:** Detailed tracking via an Electronic Data Capture (EDC) system including query resolution and eCRF locking.

11. Study Identifiers & Governance

Primary ID: {{V940-013}} **Registry Number:** {{NCT07221474}} **Posting ID:** {{30175588145881188}} **Sponsor/Lead Organization:** {{Merck}} **Study Title:** {{A Study of V940/Placebo + Pembrolizumab and Chemotherapy in Metastatic Squamous Non-Small Cell Lung Cancer (V940-013)}} **Official Regulatory Title:** {{A Phase 2, Randomized, Double-Blind, Placebo-Controlled Study of V940 in Combination With Pembrolizumab and Chemotherapy as First-Line Treatment for Participants With Metastatic Squamous NSCLC (INTERpath-013)}}

12. Clinical Framework (Oncology Specific)

Primary Condition: {{Metastatic Squamous Non-Small Cell Lung Carcinoma}} **Diagnosis Threshold:** {{Stage IV (M1a-M1c2), Treatment-Naive for Metastatic Disease}} **Medical Methodology:** {{Individualized Neoantigen Therapy (INT) + PD-1 Inhibitor + Platinum Doublet Chemotherapy}} **Optimization Target:** {{Delay of disease progression and prolonged Overall Survival}}

13. Study Procedures & Methodology

Management Pathway: {{First-line standard of care combined with investigational personalized immunotherapy}} **Education Protocol (HCP):** {{Training on handling/administration of V940 and immune-related adverse event (irAE) management}} **Education Protocol (Patient):** {{Informed consent, tissue sampling rationale, and AE reporting expectations}} **Quality Audit Mechanism:** {{Centralized independent radiological review of scans (BICR) & safety data monitoring}}

14. Observation & Follow-Up Schedule

Total Duration: {{Follow-up through May 2031}} **Onsite Visit Cadence:** {{Every 3 weeks (Q3W) during induction; Every 6 weeks (Q6W) during maintenance}} **Remote Monitoring Frequency:** {{As needed for safety follow-ups between cycles}} **Checklist Review Interval:** {{Prior to each dosing cycle administration}}

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				